$\boxed{\text{Case 6:}}$ $x=1, y\geq 2. \therefore 2x^2+5y^2 \geq 2\cdot(1)^2+5\cdot(2)^2 = 2\cdot(1)+5\cdot(4)$
$= 22 > 14$

$\boxed{\text{Case 7:}}$ $x=2, y=0 \therefore 2x^2+5y^2 = 2\cdot(2)^2+5\cdot(0)^2 = 8+0 = 8 < 14$

$\boxed{\text{Case 8:}}$ $x=2, y=1 \therefore 2x^2+5y^2 = 2\cdot(2)^2+5\cdot(1)^2 = 8+5 = 13 < 14$

$\boxed{\text{Case 9:}}$ $x=2, y\geq 2 \quad 2x^2+5y^2 \geq 2\cdot(2)^2+5\cdot(2)^2 = 8+20 = 28 > 14$

$\boxed{\text{Case 10:}}$ $x=3 \quad 2x^2+5y^2 \geq 2\cdot(3)^2+5\cdot(0)^2 = 18 > 14$

$\therefore$ There are no solutions in integers x and y to the equation $\boxed{2x^2+5y^2=14}$ $\square$

33. Prove that there are no solutions in positive integers x and y to the equation $x^4+y^4=625$.

$\boxed{\text{Solution:}}$ We shall first list all such positive integers n such that $n^4 < 625$.
$1^4=1$, $2^4=16$, $3^4=81$, $4^4=256$, $5^4=625$. It's easy to verify that for all $n\geq 5$, $n^4 \geq 625$. Now we shall verify the cases below:
WLOG, let $x \leq y$.

$\boxed{\text{Case 1:}}$ $x=1, y=1 \therefore x^4+y^4 = 1+1 = 2 < 625$

$\boxed{\text{Case 2:}}$ $x=1, y=2 \therefore x^4+y^4 = 1+16 = 17 < 625$

$\boxed{\text{Case 3:}}$ $x=1, y=3 \therefore x^4+y^4 = 1+81 = 82 < 625$

$\boxed{\text{Case 4:}}$ $x=1, y=4 \therefore x^4+y^4 = 1+256 = 257 < 625$

$\boxed{\text{Case 5:}}$ $x=1, y=5 \therefore x^4+y^4 = 1+625 = 626 > 625$

$\boxed{\text{Case 6:}}$ $x=2, y=2 \therefore x^4+y^4 = 16+16 = 32 < 625$

$\boxed{\text{Case 7:}}$ $x=2, y=3 \therefore x^4+y^4 = 16+81 = 97 < 625$